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Fir

Firs are evergreen coniferous trees belonging to the genus ***Abies*** (/ˈeɪbiˌiz/) in the family Pinaceae. There are approximately 48–65[2][3] extant species, found on mountains throughout much of North and Central America, Europe, Asia, and North Africa. The genus is most closely related to *Cedrus* (cedar). The genus name is derived from the Latin "to rise" in reference to the height of its species.[4] The common English name originates with the Old Norse, fyri, or the Old Danish, fyr.[5]

They are large trees, reaching heights of 10–80 metres (33–262 feet) tall with trunk diameters of 0.5–4 m (1 ft 8 in – 13 ft 1 in) when mature. Firs can be distinguished from other members of the pine family by the way in which their needle-like leaves are attached singly to the branches with a base resembling a suction cup, and by their cones, which, like those of true cedars, stand upright on the branches like candles and disintegrate at maturity.

Fir Temporal range: Eocene - Present[1]	
Korean fir (*Abies koreana*) cones and foliage	
Scientific classification	
Kingdom:	Plantae
Clade:	Tracheophytes
Clade:	Gymnospermae
Division:	Pinophyta
Class:	Pinopsida
Order:	Pinales
Family:	Pinaceae
Subfamily:	Abietoideae
Genus:	*Abies* Mill.
Type species	
Abies alba Mill.	
Species	
See text	
Synonyms	
▪ *Peuce* Richard 1810	

Identification of the different species is based on the size and arrangement of the leaves, the size and shape of the cones, and whether the bract scales of the cones are long and exserted, or short and hidden inside the cone.

Description

Leaves

A. alba – the underside of leaves have two whitish strips formed by wax-covered stomatal bands; the bases of the needles are shaped like suction cups.

A. grandis foliage – upper side of the leaves

Atypical *A. alba* foliage from Dinaric calcareous fir forests on Mt. Orjen

Firs can be distinguished from other members of the pine family by the unique attachment of their needle-like leaves to the twig by a base that resembles a small suction cup.

The leaves are significantly flattened, sometimes even looking like they are pressed, as in *A. sibirica*.

The leaves have two whitish lines on the bottom, each of which is formed by wax-covered stomatal bands. In most species, the upper surface of the leaves is uniformly green and shiny, without stomata or with a few on the tip, visible as whitish spots. Other species have the upper surface of leaves dull, gray-green or bluish-gray to silvery (glaucous), coated by wax with variable number of stomatal bands, and not always continuous. An example species with shiny green leaves is *A. alba*, and an example species with dull waxy leaves is *A. concolor*.

The tips of leaves are usually more or less notched (as in *A. firma*), but sometimes rounded or dull (as in *A. concolor*, *A. magnifica*) or sharp and prickly (as in *A. bracteata*, *A. cephalonica*, *A. holophylla*). The leaves of young plants are usually sharper.

The way they spread from the shoot is very diverse, only in some species comb-shaped, with the leaves arranged on two sides, flat (*A. alba*) [6]

The upper foliage is different on cone-bearing branches, with the leaves short, curved, and sharp.[7]

Cones

Intact and disintegrated Bulgarian fir cones

Immature cones of some species or races are green, not purple-bluish: for instance, Manchurian fir.

Disintegrating cones of Manchurian fir

Firs differ from other conifers in having erect, cylindrical cones 5–25 cm (2–10 in) long that disintegrate at maturity to release the winged seeds.

In contrast to spruces, fir cones do not hang. Even large fir cones grow upwards like "candles", the new growth of the tree.

Mature cones are usually brown, young in summer can be green, for example:

> *A. grandis*, *A. holophylla*, *A. nordmanniana*

or purple and blue, sometimes very dark:

> *A. fraseri*, *A. homolepis* (var. *umbellata* green), *A. koreana* ('Flava' green), *A. lasiocarpa*, *A. nephrolepis* (f. *chlorocarpa* green), *A. sibirica*, *A. veitchii* (var. *olivacea* green).[6]

Phytochemistry

Abies produce a variety of terpenoids. The analyses of the Zavarin group – from Smedman et al 1969 to Zavarin et al 1977 – showed variation in terpenoid composition of the bark by genetics, geography, age and size of the tree.[8][9]

Classification

The oldest pollen assignable to the genus dates to the Late Cretaceous in Siberia, with records of leaves and reproductive organs across the Northern Hemisphere from the Eocene onwards.[10]

Section *Abies*

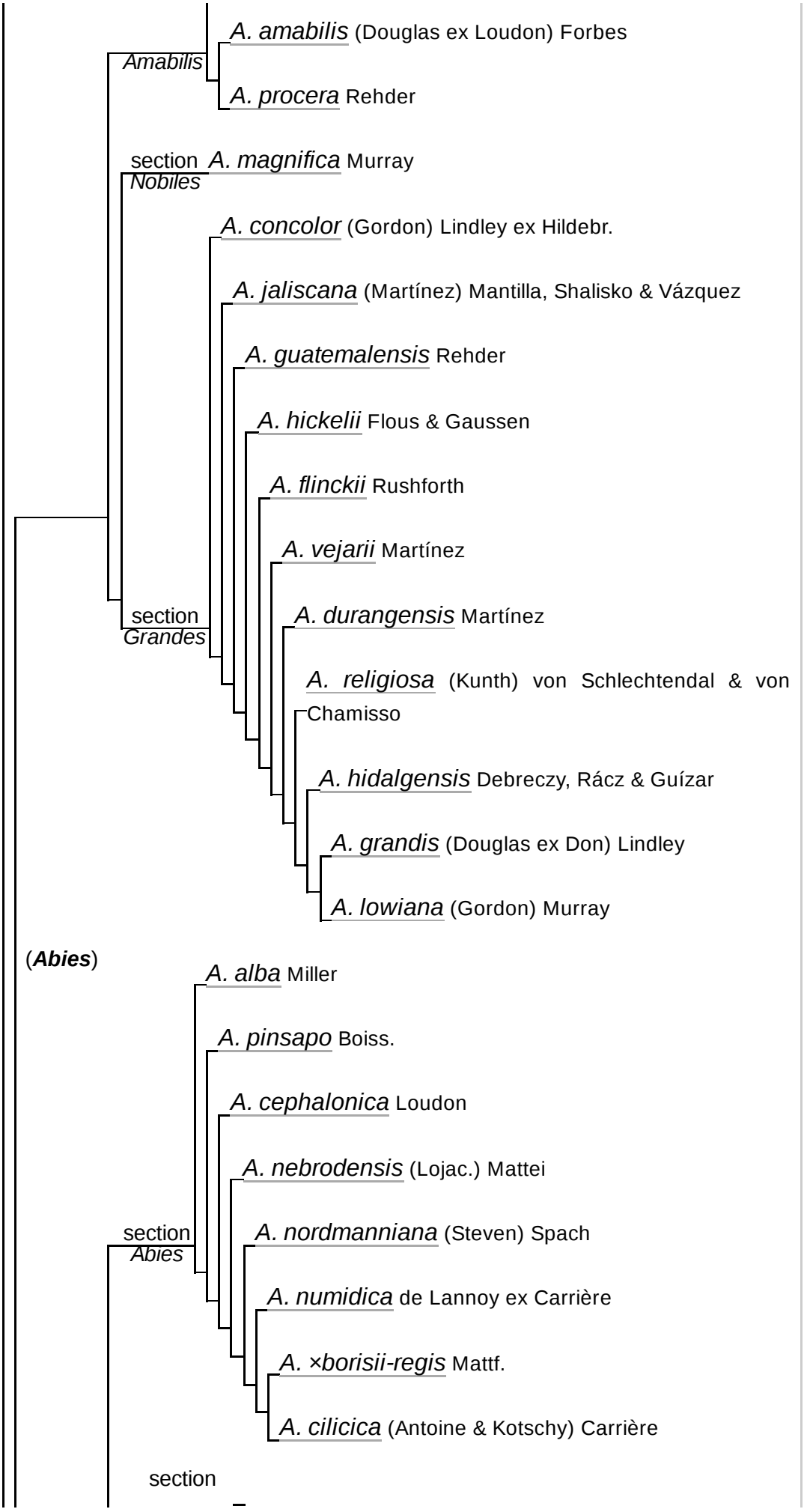
Phylogeny of *Abies*[11][12]

(***Pseudotorreya***) *A. bracteata* (Don) Poit.

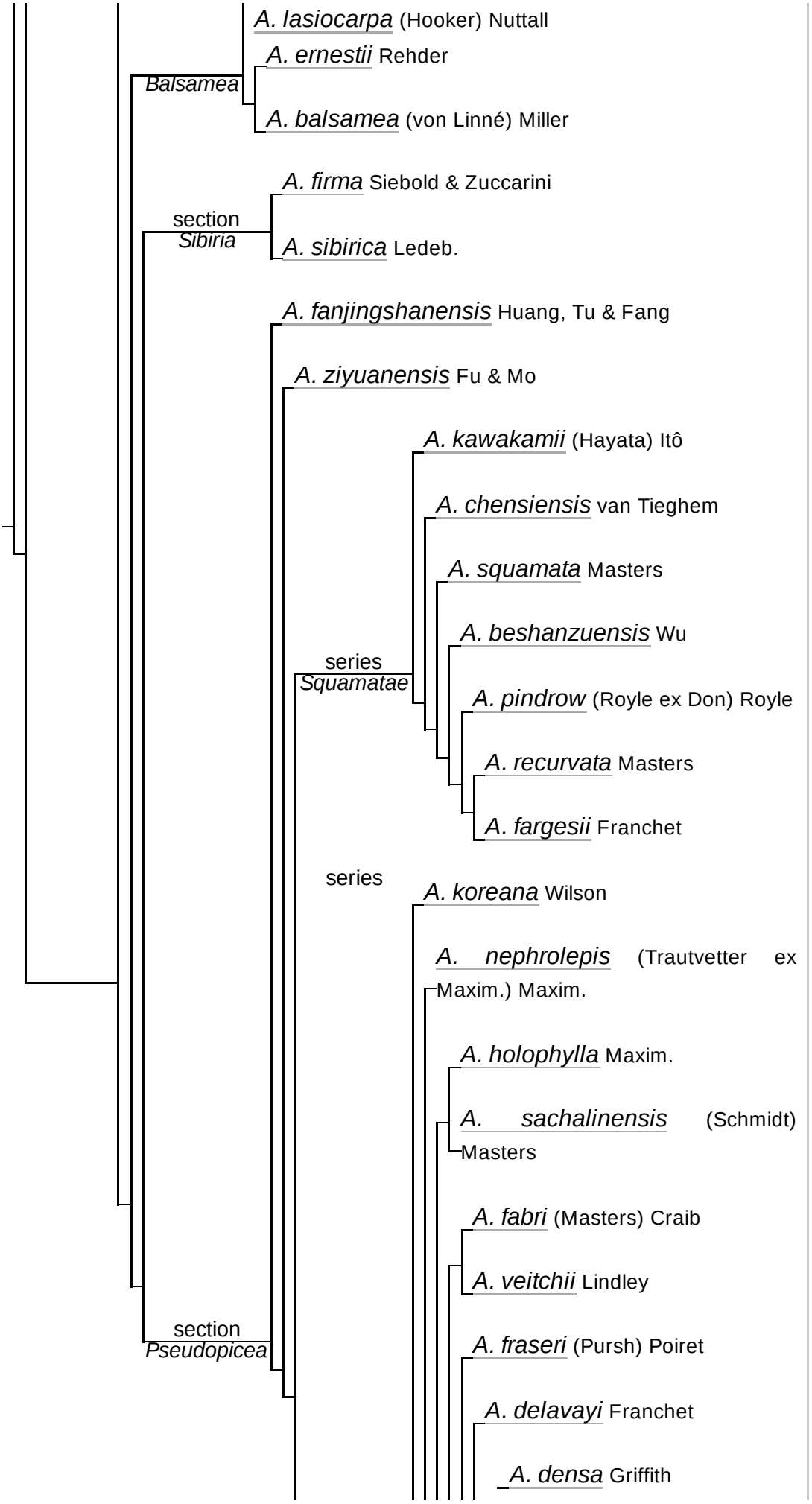
(***Religio***) section *A. mariesii* Masters

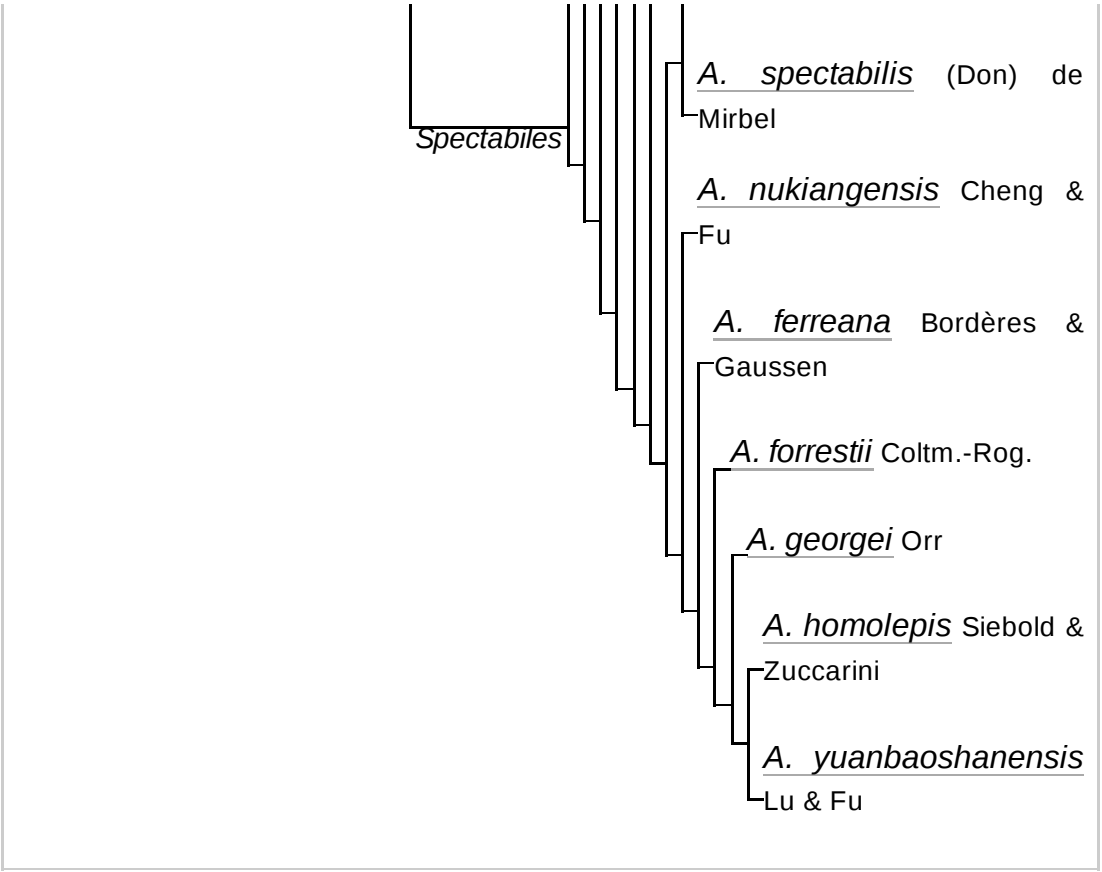
Section *Abies* is found in central, south, and eastern Europe and Asia Minor.

- *Abies alba* – silver fir
- *Abies nebrodensis* – Sicilian fir
- *Abies borisii-regis* – Bulgarian fir
- *Abies cephalonica* – Greek fir



- *Abies*


A. lasiocarpa (Hooker) Nuttall
A. ernestii Rehder
Balsamea
A. balsamea (von Linné) Miller
A. firma Siebold & Zuccarini
section Sibiria
A. sibirica Ledeb.
A. fanjingshanensis Huang, Tu & Fang
A. ziyuanensis Fu & Mo
A. kawakamii (Hayata) Itô
A. chensiensis van Tieghem
A. squamata Masters
A. beshanzuensis Wu
series Squamatae
A. pindrow (Royle ex Don) Royle
A. recurvata Masters
A. fargesii Franchet
series
A. koreana Wilson
A. nephrolepis (Trautvetter ex Maxim.) Maxim.
A. holophylla Maxim.
A. sachalinensis (Schmidt) Masters
A. fabri (Masters) Craib
A. veitchii Lindley
section Pseudopicea
A. fraseri (Pursh) Poiret
A. delavayi Franchet
A. densa Griffith



nordmanniana – Nordmann fir or Caucasian fir

- *Abies nordmanniana* subsp. *equi-trojani* – Kazdağı fir, Turkish fir

- *Abies pinsapo* – Spanish fir
  - *Abies pinsapo* var. *marocana* – Moroccan fir
- *Abies numidica* – Algerian fir
- *Abies cilicica* – Syrian fir

Section *Balsamea*

Section *Balsamea* is found in northern Asia and North America, and high mountains further south.

- *Abies fraseri* – Fraser fir
- *Abies balsamea* – balsam fir
  - *Abies balsamea* var. *phanerolepis* – bracted balsam fir
- *Abies lasiocarpa* – subalpine fir
  - *Abies lasiocarpa* var. *arizonica* – corkbark fir
  - *Abies lasiocarpa* var. *bifolia* – Rocky Mountains subalpine fir
- *Abies sibirica* – Siberian fir
  - *Abies sibirica* var. *semenovii*
- *Abies sachalinensis* – Sakhalin fir
- *Abies koreana* – Korean fir
- *Abies nephrolepis* – Khinghan fir
- *Abies veitchii* – Veitch's fir

- *Abies veitchii* var. *sikokiana* – Shikoku fir

Section *Grandis*

Section *Grandis* is found in western North America to Mexico, Guatemala, Honduras and El Salvador, in lowlands in the north, moderate altitudes in south.

- *Abies grandis* – grand fir or giant fir
  - *Abies grandis* var. *grandis* – Coast grand fir
  - *Abies grandis* var. *idahoensis* – interior grand fir
- *Abies concolor* – white fir
  - *Abies concolor* subsp. *concolor* – Rocky Mountain white fir or Colorado white fir
  - *Abies concolor* subsp. *lowiana* – Low's white fir or Sierra Nevada white fir
- *Abies durangensis* – Durango fir
  - *Abies durangensis* var. *coahuilensis* – Coahuila fir
- *Abies flinckii* – Jalisco fir
- *Abies guatemalensis* – Guatemalan fir
  - *Abies guatemalensis* var. *guatemalensis*
  - *Abies guatemalensis* var. *jaliscana*
- *Abies vejarii*

Section *Momi*

Section *Momi* is found in east and central Asia and the Himalaya, generally at low to moderate altitudes.

- *Abies kawakamii* – Taiwan fir
- *Abies homolepis* – Nikko fir
- *Abies recurvata* – Min fir
  - *Abies recurvata* var. *ernestii* – Min fir
- *Abies firma* – Momi fir
- *Abies beshanzuensis* – Baishanzu fir
- *Abies holophylla* – Manchurian fir
- *Abies chensiensis* – Shensi fir
  - *Abies chensiensis* subsp. *salouenensis* – Salween fir
- *Abies pindrow* – Pindrow fir
- *Abies ziyuanensis* – Ziyuan fir

Section *Amabilis*

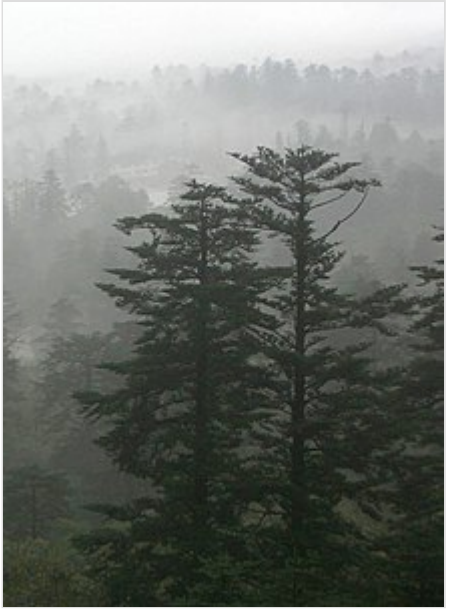
Section *Amabilis* is found in the Pacific Coast mountains in North America and Japan, in high rainfall areas.

- *Abies amabilis* – Pacific silver fir

- *Abies mariesii* – Maries' fir

Section *Pseudopicea*

Section *Pseudopicea* is found in the Sino – Himalayan mountains at high altitudes.



A. fabri, Sichuan, China

- *Abies delavayi* – Delavay's fir
  - *Abies delavayi* var. *nukiangensis*
  - *Abies delavayi* var. *motuoensis*
  - *Abies delavayi* subsp. *fansipanensis*
- *Abies fabri* – Faber's fir
  - *Abies fabri* subsp. *minensis*
- *Abies forrestii* – Forrest's fir
- *Abies densa* – Bhutan fir
- *Abies spectabilis* – East Himalayan fir
- *Abies fargesii* – Farges' fir
- *Abies fanjingshanensis* – Fanjingshan fir
- *Abies yuanbaoshanensis* – Yuanbaoshan fir
- *Abies squamata* – flaky fir

Section *Oiamel*

Section *Oiamel* is found in central Mexico at high altitudes.

- *Abies religiosa* – sacred fir
- *Abies hickelii* – Hickel's fir
  - *Abies hickelii* var. *oaxacana* – Oaxaca fir

Section *Nobilis*

Section *Nobilis* (western U.S., high altitudes)

- *Abies procera* – noble fir
- *Abies magnifica* – red fir
  - *Abies magnifica* var. *shastensis* – Shasta red fir

Section *Bracteata*

Section *Bracteata* (California coast)

- *Abies bracteata* – bristlecone fir

- ?†*Abies rigida* Frank Knowlton[13][14] - (Priabonian-Chattian; Colorado)

Section *Incertae sedis*

Section *Incertae sedis*

- †*Abies milleri* – (Extinct) Early Eocene[1]

A. magnifica, California, USA

Ecology

Firs are used as food plants by the caterpillars of some Lepidoptera species, including *Chionodes abella* (recorded on white fir), autumnal moth, conifer swift (a pest of balsam fir), the engrailed, grey pug, mottled umber, pine beauty and the tortrix moths *Cydia illutana* (whose caterpillars are recorded to feed on European silver fir cone scales) and *C. duplicana* (on European silver fir bark around injuries or canker).

Uses

Wood of most firs is considered unsuitable for general timber use and is often used as pulp or for the manufacture of plywood and rough timber. Because this genus has no insect or decay resistance qualities after logging, it is generally recommended in construction purposes for indoor use only (e.g. indoor drywall on framing). Firwood left outside cannot be expected to last more than 12 to 18 months, depending on the type of climate it is exposed to.

Green fir twig pictured on top of heart in the coat of arms of Laukaa

Nordmann fir, noble fir, Fraser fir and balsam fir are popular Christmas trees, generally considered to be the best for this purpose, with aromatic foliage that does not shed many needles on drying out. Many are also decorative garden trees, notably Korean fir and Fraser fir, which produce brightly coloured cones even when very young, still only 1–2 m (3 ft 3 in – 6 ft 7 in) tall. Other firs can grow anywhere between 30 and 236 feet (9.1 and 71.9 m) tall. Many fir species are grown in botanic gardens and other specialist tree collections in Europe and North America.[15] Fir Tree Appreciation Day is June 18.

Abies religiosa—sacred fir, is the overwinter host for the monarch butterfly.[16][17]

Abies spectabilis or Talispatra is used in Ayurveda as an antitussive (cough suppressant) drug.[18][19]

See also

- Douglas fir
- List of fir species of Denmark

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Further reading

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External links

- *Abies* (http://www.conifers.org/pi/Abies.php) at The Gymnosperm Database
- *Abies* (https://treesandshrubsonline.org/articles/abies/) from the website *Trees and Shrubs Online*
- Michael P. FRANKIS CONE COLLECTION: Abies (http://www.pinetum.org/cones/ABIEScones.htm) at the Arboretum de Villardebelle—images of cones of selected species
- Platt, Karen (http://www.karenplatt.co.uk) "Gold Fever" provides descriptions of golden or yellow-leaved Abies cultivars

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